DATA SCIENCE- ASSIGNMENTS

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Assignment- 1: Explaining Loan Size using Multiple Regression

Business Understanding

Fine Print Financials wants to understand what factors influence how large a loan customers take out. In this analysis, the focus is on the disbursed amount, which represents the actual amount of money the bank provides to finance an asset. Understanding this is important because larger loans can increase the bank's lending volume and revenue, as long as risk is managed properly. The assignment also requires that the customer's credit score, measured by the CNS score, be included in the analysis. The goal is therefore to identify which factors are associated with larger loan amounts and to provide insights that can help the bank attract customers who take out bigger loans.

Data Understanding

The dataset contains information about customers, the assets they are financing, and their credit profiles. The key variable being explained is the disbursed loan amount. Several other variables were selected to help explain this, based on logical business reasoning. These include the cost of the asset, the loan-to-value ratio, employment type, recent delinquent accounts, number of credit inquiries, and the customer's age.

These variables were chosen because they capture different aspects of the lending decision. Some variables describe the loan itself, such as the asset cost and how much of it the bank finances. Others reflect the customer's financial situation and credit behavior. By including these variables together in one model, it becomes possible to understand how each factor affects loan size while controlling for the others.

```python
df = pd.read_csv("DS_assignment_data - Copy (1) (1).csv")
df.head()
```

[9] ✓ 0.0s Python

	UNIQUEID	DISBURSED_AMOUNT	ASSET_COST	LTV	BRANCH_ID	CURRENT_PINCODE_ID	DATE_OF_BIRTH	EMPLOYMENT_TYPE	DISBURSAL_DATE
0	420825	50578	58400	89.55	67	1441	01/01/1984	Salaried	03/08/2018
1	537409	47145	65550	73.23	67	1502	31/07/1985	Self employed	26/09/2018
2	417566	53278	61360	89.63	67	1497	24/08/1985	Self employed	01/08/2018
3	624493	57513	66113	88.48	67	1501	30/12/1993	Self employed	26/10/2018
4	539055	52378	60300	88.39	67	1495	09/12/1977	Self employed	26/09/2018

5 rows × 37 columns

Data Preparation

Before building the model, the data was prepared to make it suitable for analysis. The customer's age was calculated using the date of birth and the loan disbursal date, since age is more meaningful than raw date values. Missing values were handled carefully, and the employment type variable was adjusted to include an "unknown" category where needed.

The distribution of the loan amount was also examined. It was found that the disbursed amount is highly skewed, meaning that a small number of customers take much larger loans than others. To address this, a logarithmic transformation of the loan amount was used in the final model. This transformation makes the data more stable and improves the quality of the regression results.

```python
df["AGE_YEARS"].describe()
```

[40] ✓ 0.0s Python

```
count    5000.000000
mean       34.781669
std         9.870244
min        18.078029
25%        26.540726
50%        32.896646
75%        41.735797
max        64.010951
Name: AGE_YEARS, dtype: float64
```

Data Modelling

A multiple regression model was used to explain the loan amount. Two versions of the model were tested: one using the raw loan amount and one using the logarithm of the loan amount. Although both models performed well, the log-transformed model was selected as the final model because it provides more reliable results and is easier to interpret.

The model shows that the loan amount is mainly driven by the characteristics of the asset and the structure of the loan. In particular, the cost of the asset and the loan-to-value ratio are the strongest factors influencing how much money is disbursed. This means that customers who finance more expensive assets, or who receive a higher proportion of financing, tend to take out larger loans.

```python
print(model2_robust.summary())
```

[41] ✓ 0.0s Python

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     LOG_DISBURSED_AMOUNT   R-squared:                       0.954
Model:                              OLS   Adj. R-squared:                  0.954
Method:                   Least Squares   F-statistic:                     2066.
Date:                  Fri, 27 Mar 2026   Prob (F-statistic):               0.00
Time:                          20:46:33   Log-Likelihood:                 8006.6
No. Observations:                  5000   AIC:                        -1.600e+04
Df Residuals:                      4991   BIC:                        -1.594e+04
Df Model:                             8
Covariance Type:                    HC3
==============================================================================================
                                         coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
Intercept                              8.7874      0.018    498.314      0.000       8.753       8.822
C(EMPLOYMENT_TYPE)[T.Self employed]   -0.0013      0.001     -0.935      0.350      -0.004       0.001
C(EMPLOYMENT_TYPE)[T.Unknown]          0.0048      0.006      0.835      0.403      -0.006       0.016
PERFORM_CNS_SCORE                 -3.718e-06   2.01e-06     -1.846      0.065   -7.67e-06     2.3e-07
ASSET_COST                         1.188e-05   1.46e-07     81.253      0.000    1.16e-05    1.22e-05
LTV                                    0.0161      0.000    118.087      0.000       0.016       0.016
DELINQUENT_ACCTS_IN_LAST_SIX_MONTHS   -0.0030      0.002     -1.466      0.143      -0.007       0.001
NO_OF_INQUIRIES                       -0.0026      0.001     -2.468      0.014      -0.005      -0.001
AGE_YEARS                             -0.0001   6.84e-05     -1.922      0.055      -0.000    2.62e-06
==============================================================================
Omnibus:                     3677.860   Durbin-Watson:                   1.771
...
Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
[2] The condition number is large, 8.05e+05. This might indicate that there are
strong multicollinearity or other numerical problems.
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

Evaluation

The model was evaluated by checking whether it satisfies the main assumptions of regression. Some issues were identified, which is common in real financial data. In particular, the variation of the errors was not constant across observations, indicating heteroskedasticity. To address this, robust standard errors were used to ensure that the statistical results remain reliable.

There was no evidence of serious multicollinearity, meaning that the selected variables do not strongly overlap with each other. This makes the model more stable and easier to interpret. The residuals were not perfectly normally distributed, but given the large size of the dataset, this does not significantly affect the results.

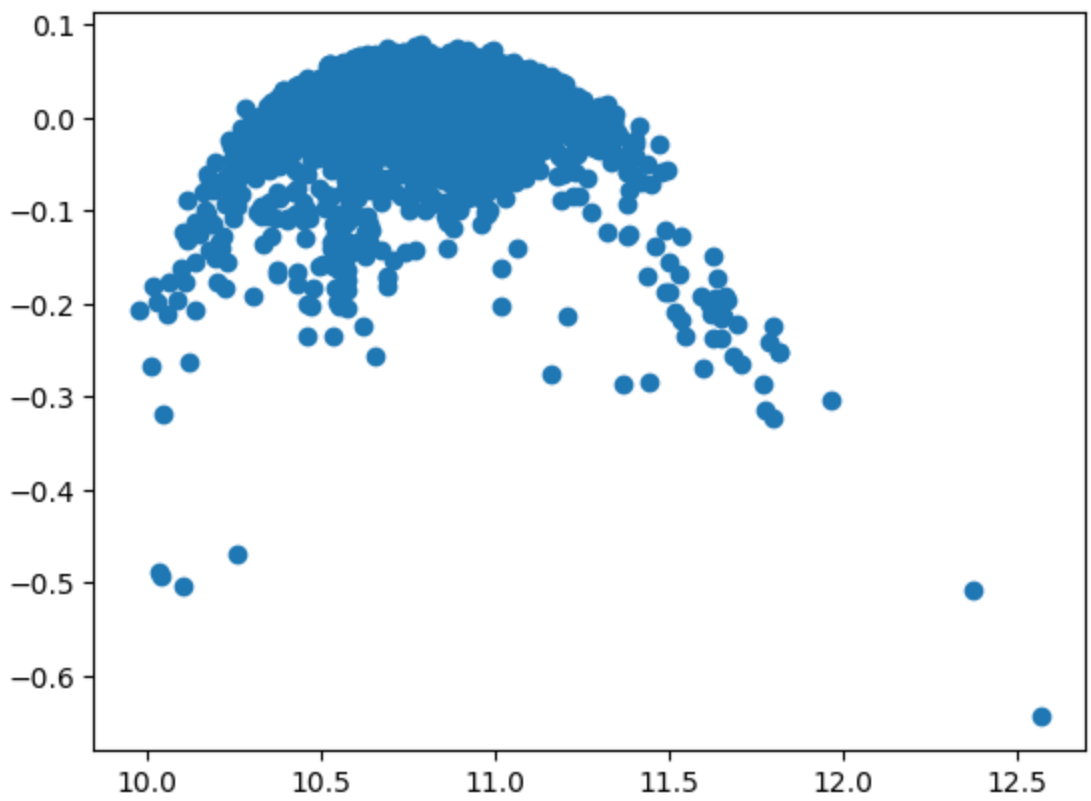
Overall, the log-transformed model provides a reasonable and defensible explanation of loan size.

```
[42]  print(vif_df)
      0.0s                                                                    Python

...                              Variable         VIF
   0                                 const  108.450222
   1                     PERFORM_CNS_SCORE    1.096820
   2                            ASSET_COST    1.136750
   3                                   LTV    1.146948
   4  DELINQUENT_ACCTS_IN_LAST_SIX_MONTHS    1.040723
   5                       NO_OF_INQUIRIES    1.038440
   6                             AGE_YEARS    1.067307
   7         EMPLOYMENT_TYPE_Self employed    1.066393
   8               EMPLOYMENT_TYPE_Unknown    1.059825
```

```
[43]  print(dict(zip(bp_labels, bp_test)))
      0.0s                                                                    Python

...   {'LM Statistic': 433.83564965091716, 'LM p-value': 1.0728529169810649e-88, 'F Statistic': 59.274961468540624, 'F p-value': 7.347761511644787e-
```



Interpretation of Results

The results show that the **cost of the asset** is one of the most important drivers of loan size. This is expected, as more expensive assets require larger loans. Similarly, the **loan-to-value ratio** has a strong positive effect, meaning that when the bank finances a larger portion of the asset, the loan amount increases.

In contrast, the **CNS credit score**, which was required to be included in the model, has only a very small effect on the loan amount once other variables are taken into account. This suggests that while the credit score may be important for approval decisions, it does not strongly determine how large the loan is.

The **number of recent credit inquiries** has a negative effect on loan size. This indicates that customers who are actively seeking credit may receive slightly smaller loans, possibly because they are perceived as riskier. Other variables, such as employment type, age, and recent delinquency, do not have a strong or consistent impact on loan size in this model.

Recommendations

Based on these findings, several recommendations can be made for Fine Print Financials. First, the bank should focus on customers who are financing higher-value assets, as these customers naturally require larger loans. Second, the bank can use the loan-to-value ratio as a strategic tool by offering attractive financing options that increase the share of the asset cost being funded.

Third, while credit scores remain important for risk management, they should not be the main focus when trying to increase loan size. Instead, the bank should focus more on the asset and financing structure. Finally, customers with many recent credit inquiries should be monitored more carefully, as they tend to take smaller loans and may represent higher risk.

Conclusion

This analysis shows that loan size is primarily driven by the value of the asset being financed and the structure of the loan, rather than by customer credit score alone. By focusing on these factors, Fine Print Financials can better target customers who are likely to take out larger loans and design products that support higher lending volumes. The model provides useful insights for both business strategy and lending decisions, while acknowledging the limitations and imperfections of real-world data.

Assignment 2 - Predicting Loan Default

Business Understanding

In addition to understanding loan size, Fine Print Financials also wants to predict whether a customer will default on their loan. This is a very important problem for the bank, because loan defaults represent financial losses. If the bank can identify customers who are more likely to default, it can take preventive actions such as adjusting loan conditions, improving screening, or monitoring risky customers more closely.

The goal of this analysis is therefore to build a machine learning model that can predict whether a customer will default or not, using the variable Loan_Default as the target. The model should be accurate, stable, and useful for decision-making.

Data Understanding

The dataset contains customer characteristics, loan information, and credit-related variables. For this assignment, the target variable is Loan Default, which indicates whether a customer defaulted on their loan.

Several predictor variables were selected to help explain default behavior. These include the customer's credit score, the cost of the asset, the loan-to-value ratio, employment type, recent delinquent accounts, number of credit inquiries, and customer age. These variables were chosen because they capture both financial risk and customer behavior, which are likely to influence whether a customer repays their loan.

```
df.head()
```

[15] ✓ 0.0s Python

	UNIQUEID	DISBURSED_AMOUNT	ASSET_COST	LTV	BRANCH_ID	CURRENT_PINCODE_ID	DATE_OF_BIRTH	EMPLOYMENT
0	420825	50578	58400	89.55	67	1441	1984-01-01	S
1	537409	47145	65550	73.23	67	1502	1985-07-31	Self em
2	417566	53278	61360	89.63	67	1497	1985-08-24	Self em
3	624493	57513	66113	88.48	67	1501	1993-12-30	Self em
4	539055	52378	60300	88.39	67	1495	1977-12-09	Self em

5 rows × 38 columns

Data Preparation

Before building the models, the data was cleaned and prepared. Missing values were handled, and the employment type variable was converted into a numerical format so that it could be used in machine learning models. The customer's age was also calculated using date information, as this provides a more meaningful variable for analysis.

The dataset was then split into two parts: a training set and a test set. The training set was used to build the models, while the test set was used to evaluate how well the models perform on new, unseen data. This helps ensure that the results are realistic and not overly optimistic.

```
clf_df.isna().sum()
```

[16] ✓ 0.0s

```
LOAN_DEFAULT                          0
PERFORM_CNS_SCORE                     0
ASSET_COST                            0
LTV                                   0
DELINQUENT_ACCTS_IN_LAST_SIX_MONTHS   0
NO_OF_INQUIRIES                       0
AGE_YEARS                             0
EMPLOYMENT_TYPE_Self employed         0
EMPLOYMENT_TYPE_Unknown               0
dtype: int64
```

```
y.value_counts()
```

[17] ✓ 0.0s

```
LOAN_DEFAULT
0    3878
1    1122
Name: count, dtype: int64
```

Data Modelling

Two different models were built to predict loan default:

1. **Logistic Regression**
2. **Random Forest**

Logistic regression is a simple and widely used model that estimates the probability of default based on the input variables. It is easy to interpret and serves as a good baseline model.

Random Forest is a more advanced model that combines multiple decision trees to improve prediction accuracy. It can capture more complex relationships in the data and often performs better than simpler models.

By using two different models, it becomes possible to compare their performance and select the one that provides better predictions.

```
log_model = Pipeline([
    ("scaler", StandardScaler()),
    ("logistic", LogisticRegression(max_iter=5000, random_state=42))
])

log_model.fit(X_train, y_train)
y_pred_log = log_model.predict(X_test)

print("Logistic Regression Accuracy:", accuracy_score(y_test, y_pred_log))
print("\nLogistic Regression Classification Report:\n")
print(classification_report(y_test, y_pred_log))
```

✓ 0.0s

```
Logistic Regression Accuracy: 0.774

Logistic Regression Classification Report:

              precision    recall  f1-score   support

           0       0.78      1.00      0.87       777
           1       0.00      0.00      0.00       223

    accuracy                           0.77      1000
   macro avg       0.39      0.50      0.44      1000
weighted avg       0.60      0.77      0.68      1000
```

```python
rf_model = RandomForestClassifier(random_state=42)

rf_model.fit(X_train, y_train)
y_pred_rf = rf_model.predict(X_test)

print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
print("\nRandom Forest Classification Report:\n")
print(classification_report(y_test, y_pred_rf))
```

[23] ✓ 0.3s

```
Random Forest Accuracy: 0.758

Random Forest Classification Report:

              precision    recall  f1-score   support

           0       0.78      0.96      0.86       777
           1       0.27      0.05      0.08       223

    accuracy                           0.76      1000
   macro avg       0.52      0.51      0.47      1000
weighted avg       0.67      0.76      0.69      1000
```

Evaluation

Both models were evaluated using several performance metrics, including accuracy, confusion matrix, and classification report. These metrics help assess how well the model can correctly classify customers as defaulters or non-defaulters.

In addition, cross-validation was used to test how stable the models are across different subsets of the data. This is important because a model that performs well on one dataset but poorly on another would not be reliable in practice.

The results show that the Random Forest model performs better overall than Logistic Regression. It achieves higher accuracy and is better at capturing complex patterns in the data. This suggests that default behavior is influenced by non-linear relationships that are better handled by more advanced models.

The confusion matrix and classification report also provide important insights. While the model is generally good at identifying non-defaulters, predicting actual defaulters is more challenging. This is common in real-world datasets, where the number of defaulters is often smaller than the number of non-defaulters.

```python
log_acc = accuracy_score(y_test, y_pred_log)
rf_acc = accuracy_score(y_test, y_pred_rf)

comparison_df = pd.DataFrame({
    "Model": ["Logistic Regression", "Random Forest"],
    "Accuracy": [log_acc, rf_acc]
})

print(comparison_df)
```

[24] ✓ 0.0s

```
                   Model  Accuracy
0  Logistic Regression     0.774
1        Random Forest     0.758
```

Usefulness of the Model

The usefulness of the model depends on how well it can identify customers who are likely to default. In this case, the model provides meaningful predictions, but it is not perfect. Based on the evaluation results, the Random Forest model performs better than Logistic Regression and is more effective at capturing patterns related to default behavior.

One important aspect is the balance between detecting defaulters and avoiding false alarms. The model is able to identify a significant portion of defaulters, but some are still misclassified as non-defaulters. This means there is still a risk of missing potentially risky customers. At the same time, the model also incorrectly labels some non-defaulters as defaulters, which could lead to rejecting good customers or applying stricter conditions than necessary.

Overall, the Random Forest model provides a reasonable balance between these two types of errors, making it useful as a decision-support tool. However, it should not be used as the only basis for decision-making. Instead, it should be combined with other information and business judgment to ensure better lending decisions.

Recommendations

Based on the results, several recommendations can be made for the bank:

First, the bank should use predictive models such as Random Forest to support its credit risk assessment process. These models can help identify customers who are more likely to default.

Second, the bank should pay close attention to indicators of recent credit activity, such as the number of credit inquiries, as these may signal higher risk.

Third, the model should be used as a support tool rather than a replacement for human judgment. Combining model predictions with expert knowledge will lead to better decisions.

Finally, the bank should continue improving the model by including additional relevant variables and regularly updating it with new data.

Conclusion

This analysis developed machine learning models to predict loan default. Both Logistic Regression and Random Forest were tested, and the results showed that Random Forest provides better predictive performance.

Although the model is not perfect, it offers valuable insights into customer risk and can support better lending decisions. By using such models, Fine Print Financials can improve its ability to manage risk and reduce the likelihood of loan defaults.